Fluids and Electrolytes

Fluid management • Osmolality • Hypertonic saline • Hypotonic fluids • Sodium • Potassium • Magnesium • Phosphorus • Calcium — all key points for the exam, with drug tables

1. Body Water and Fluid Management

Body water compartments

- **Total body water (TBW)** = **60%** of body weight in adult men, **50%** in adult women (elderly: men 50%, women 45%; infants up to 75–80%).
- **2/3 intracellular (ICF)**, **1/3 extracellular (ECF)** → ECF = **3/4 interstitial** + **1/4 intravascular (plasma)**.
- Example (70 kg man): TBW 42 L → ICF 28 L, ECF 14 L (interstitial 10.5 L, plasma 3.5 L).

Maintenance requirements (adults)

Item	Requirement
Water	**30–35 mL/kg/day** (≈ 1 mL per kcal) • **Holliday-Segar (4-2-1 rule)**: 4 mL/kg/h for first 10 kg + 2 mL/kg/h for next 10 kg + 1 mL/kg/h for each kg > 20 kg (70 kg → 110 mL/h) • daily form: 100 / 50 / 20 mL/kg/day
Sodium	**1–2 mEq/kg/day**
Potassium	**0.5–1 mEq/kg/day**
Dextrose	~100–150 g/day prevents ketosis/protein breakdown (D5 provides 50 g/L = 170 kcal/L) • **not nutrition**

Crystalloid composition (high-yield)

Fluid	Na	Cl	K	Other	mOsm/L	Tonicity / notes
0.9% NaCl (normal saline)	154	154	0	—	308	**Isotonic** • large volumes → **hyperchloremic metabolic acidosis**, AKI
Lactated Ringer's	130	109	4	Ca 3, lactate 28	273	Isotonic/balanced • lactate → bicarbonate (liver) • contains **calcium** (don't mix with blood or ceftriaxone in neonates)
Plasma-Lyte A	140	98	5	Mg 3, acetate 27, gluconate 23	294	Isotonic/balanced • pH 7.4
D5W	0	0	0	Dextrose 50 g/L (170 kcal)	252	Dextrose is metabolized → acts as **free water** (hypotonic in the body)
0.45% NaCl (½ NS)	77	77	0	—	154	**Hypotonic**
D5 ½NS ± 20 KCl	77	77	0–20	Dextrose 50 g/L	406–446	Classic maintenance (hypotonic in body)
3% NaCl	513	513	0	—	1,026	**Hypertonic**
23.4% NaCl	4,000 (4 mEq /mL)	4,000	0	—	~8,000	**Concentrated electrolyte** — must be diluted • central line • ICP

Distribution of 1 liter infused

Fluid	Stays intravascular	Use
Isotonic crystalloid (NS, LR, Plasma-Lyte)	**~250 mL** (¼; distributes through ECF)	**Resuscitation**, ECF replacement
D5W	**~80 mL** (1/12; distributes through TBW)	**Free water** replacement (hypernatremia) • never for resuscitation
Albumin 5%	Most (initially)	Volume expansion (no survival benefit vs crystalloids)
Albumin 25%	**Pulls ~3–4× its volume** from interstitium	Cirrhosis (large-volume paracentesis, SBP, HRS), hypoalbuminemia with edema

Colloids

- **Albumin 5%** (iso-oncotic) and **25%** (hyperoncotic) • expensive • no outcome benefit over crystalloids for general resuscitation (SAFE trial) • avoid albumin in **traumatic brain injury** (↑ mortality).
- **Hydroxyethyl starch (HES)**: **boxed warning** — ↑ **mortality**, **AKI/renal replacement therapy**, and **bleeding** → **avoid in critically ill, sepsis**.
- **Dextran**: anaphylaxis, bleeding (impairs platelets), AKI — rarely used.

Choosing resuscitation fluid

- **Balanced crystalloids** (LR, Plasma-Lyte) are preferred for most critically ill patients (↓ major adverse kidney events vs saline — SMART trial).
- **Normal saline preferred**: **traumatic brain injury** (LR is slightly hypotonic → cerebral edema), **hypochloremic metabolic alkalosis** (vomiting), hyponatremia.
- **Sepsis/septic shock: 30 mL/kg crystalloid within the first 3 hours**, then reassess (dynamic measures, passive leg raise, lactate).
- **Hemorrhage:** blood products (balanced 1:1:1) • limit crystalloids.
- **Fluid overload** is harmful (pulmonary edema, AKI, ↑ mortality) → reassess frequently; de-resuscitate with diuretics when stable.

Assessing volume status

Hypovolemia	Hypervolemia
Tachycardia, **hypotension**, **orthostatic** changes, dry mucous membranes, ↓ skin turgor, ↓ urine output (< 0.5 mL/kg/h), weight loss, **BUN:SCr > 20**, ↑ hematocrit, concentrated urine (low urine Na)	**Edema**, **crackles**, ↑ JVP, weight gain, S3, pleural effusion, ascites, ↓ oxygenation

Maintenance fluids in hospitalized patients

- Use **isotonic fluids** (with dextrose and potassium as needed) for maintenance in most hospitalized patients — **hypotonic maintenance fluids** ↑ risk of **hospital-acquired hyponatremia** (AAP 2018 guideline for children).
- Reassess daily • stop IV fluids when oral intake is adequate.

2. Osmolality and Tonicity

- **Osmolality** = osmoles per kg of water (measured) • **osmolarity** = per liter of solution (calculated).
- **Normal serum osmolality: 275–295 mOsm/kg**.
- **Calculated serum osmolality** = **2 × Na + glucose/18 + BUN/2.8** (mg/dL units).
- **Effective osmolality (tonicity)** = 2 × Na + glucose/18 → **urea is an ineffective osmole** (crosses cell membranes freely).
- **Osmolal gap** = measured − calculated • normal **< 10** • **↑ gap** → unmeasured osmoles: **methanol**, **ethylene glycol**, ethanol, isopropyl alcohol, **mannitol**, **propylene glycol** (vehicle in IV **lorazepam**, diazepam, phenytoin), glycine.
- **Sodium correction for hyperglycemia:** corrected Na = measured Na + **1.6 × [(glucose − 100)/100]** (use 2.4 if glucose > 400).

- **Tonicity effects on cells:** hypotonic fluid → water moves **into** cells (swelling, cerebral edema) • hypertonic → water moves **out** of cells (shrinking).
- **Urine osmolality** 50–1,200 mOsm/kg (ADH-dependent): < 100 → dilute (ADH suppressed: polydipsia, low solute) • > 100 → ADH acting (e.g., SIADH).
- **IV solution osmolarity** matters: **peripheral veins tolerate ≤ 900 mOsm/L** (e.g., peripheral PN) → higher needs a central line.

3. Hypertonic Saline

Product	Concentration	Uses / key points
3% NaCl	**513 mEq/L** Na	**Severe/symptomatic hyponatremia**: **100–150 mL IV bolus over 10–20 min**, repeat up to 2–3 times until symptoms improve or Na ↑ 4–6 mEq/L • slow infusion for correction • **cerebral edema/↑ ICP** • can be given peripherally (short term)
23.4% NaCl	**4 mEq/mL**	**Raised intracranial pressure** (e.g., 30 mL over 10–20 min via **central line**) • additive for PN/IV bags • **high-alert medication** (ISMP) — store separately, never give undiluted peripherally

- **Risks: overcorrection → osmotic demyelination syndrome (ODS)** • volume overload/pulmonary edema • hypernatremia • hyperchloremic acidosis • phlebitis/tissue injury.
- **Monitor serum Na every 2–4 hours** during active correction • urine output (a water diuresis can cause rapid overcorrection).

Sodium formulas (calculations)

- **Change in serum Na per 1 L of infusate (Adrogué-Madias)** = **(infusate Na + infusate K − serum Na) ÷ (TBW + 1)**.
- **Example:** 70 kg man (TBW 42 L), Na 115, 1 L of 3% NaCl → (513 − 115) ÷ 43 = **+9.3 mEq/L** per liter.
- **Sodium deficit (mEq)** = TBW × (desired Na − current Na).
- **Free water deficit (L)** = TBW × [(serum Na ÷ 140) − 1] (used for hypernatremia).

4. Hypotonic Intravenous Fluids

- **Examples: D5W** (hypotonic once dextrose is metabolized), **0.45% NaCl**, 0.2% NaCl, D5 ½NS.
- **Uses: free water replacement** in **hypernatremia** • maintenance (with caution) • D5W as a diluent for drugs.
- **Risks: hospital-acquired hyponatremia** and **cerebral edema** — especially in **children**, **postoperative** patients, and anyone with **non-osmotic ADH release** (pain, nausea, vomiting, stress, opioids, surgery, CNS/lung disease) • hyperglycemia (dextrose).
- **Never use hypotonic fluids for resuscitation** (stay mostly outside the vascular space) or in **brain injury**.
- **Sterile water for injection must NEVER be infused IV alone → hemolysis** (very hypotonic) → only as a diluent; free water can be given **enterally** instead.

5. Hyponatremia and Hypo-osmolality States

Definition and symptoms

- **Na < 135 mEq/L** • mild 130–134 • moderate 125–129 • **severe < 125** • **acute** (< 48 h) vs **chronic** (> 48 h or unknown).
- **Symptoms** (cerebral edema): nausea, headache, confusion, lethargy, gait problems, **falls → severe**: vomiting, **seizures**, obtundation, **coma**, respiratory arrest • acute and severe hyponatremia cause more symptoms.

Diagnostic approach

1. **Check serum osmolality:**

2. – **Normal (isotonic): pseudohyponatremia** (severe hyperlipidemia, hyperproteinemia — e.g., multiple myeloma).
3. – **High (hypertonic): hyperglycemia** (correct Na), **mannitol**, IVIG (maltose).
4. – **Low (hypotonic, < 275):** true hyponatremia → assess **volume status**.
5. **Assess volume status** and **urine sodium/osmolality**.

Volume status	Causes	Urine Na	Treatment
Hypovolemic	**Renal losses: thiazide diuretics** (most common drug cause), mineralocorticoid deficiency (adrenal insufficiency), salt-wasting nephropathy, cerebral salt wasting • **Extrarenal:** vomiting, diarrhea, burns, third spacing	**> 20** (renal) • **< 20** (extrarenal)	**0.9% NaCl** (restores volume → ADH falls) • stop thiazide • fludrocortisone/hydrocortisone if adrenal
Euvolemic	**SIADH** (most common), **hypothyroidism**, **glucocorticoid deficiency**, **psychogenic polydipsia**, **low solute intake** (beer potomania, "tea and toast"), exercise-associated	Usually **> 30–40** (SIADH) • low in polydipsia/low solute	**Fluid restriction**, treat cause, ↑ solute, vaptans (see below)
Hypervolemic	**Heart failure**, **cirrhosis**, **nephrotic syndrome**, advanced CKD/AKI	**< 20** (unless diuretics or renal failure)	**Fluid and sodium restriction**, **loop diuretics**, treat cause • vaptans (avoid tolvaptan in cirrhosis)

SIADH

- **Criteria:** hypotonic hyponatremia, **euvolemic**, **urine osmolality > 100** (inappropriately concentrated), **urine Na > 30–40**, normal thyroid and adrenal function, low uric acid and BUN.
- **Causes:** CNS disorders (stroke, trauma, infection), **lung disease** (pneumonia, **small cell lung cancer**), **pain, nausea, surgery**, HIV.
- **Drug causes (high-yield): SSRIs**, SNRIs, TCAs, mirtazapine (elderly), **carbamazepine**, **oxcarbazepine**, antipsychotics, **cyclophosphamide**, **vincristine**, cisplatin, **chlorpropamide**, opioids, **MDMA (ecstasy)**, NSAIDs (potentiate ADH), **desmopressin**, oxytocin, PPIs (rare), valproate, lamotrigine.
- **Treatment:** treat cause, stop offending drug • **fluid restriction** (< 800–1,000 mL/day; first-line) • ↑ solute: **oral urea** (15–60 g/day), **salt tablets + loop diuretic** • **vaptans** (tolvaptan, conivaptan) • demeclocycline (nephrotoxic — rarely used).

Vasopressin receptor antagonists (vaptans)

Drug	Key points
Tolvaptan (oral, V2)	**15 mg daily** → 30–60 mg • **start in hospital** (monitor Na closely) • **boxed: too-rapid correction → ODS** • **avoid in liver disease** (hepatotoxicity; limit to **30 days**) • **do not fluid restrict** in the first 24 h • thirst, polyuria, dry mouth • **CI:** hypovolemic hyponatremia, need for urgent Na rise, anuria, **strong CYP3A4 inhibitors**
Conivaptan (IV, V1a/V2)	**20 mg IV load over 30 min** → 20 mg/day continuous infusion (max 40 mg/day) for up to **4 days** • euvolemic/hypervolemic hyponatremia in hospital • **infusion-site reactions** (change site every 24 h) • **strong CYP3A4 inhibitor** (CI with other strong 3A4 inhibitors) • hypotension

Treatment by severity and correction limits

Situation	Treatment
Severe symptoms (seizures, coma, obtundation, respiratory distress)	**3% NaCl 100–150 mL IV over 10–20 min**, repeat up to 2–3 times → goal: ↑ Na **4–6 mEq/L** (enough to reverse cerebral edema) • then slow correction
Moderate symptoms	Single 3% bolus or slow 3% infusion + treat cause
Mild/asymptomatic, chronic	Treat the cause by volume status (fluid restriction, NS, diuretics, stop drugs)

Correction limits (chronic hyponatremia): aim for **4–8 mEq/L per 24 h** • do **NOT exceed 8–10 mEq/L in 24 h** (or 18 mEq/L in 48 h) • use **≤ 8 mEq/L/24 h** in **high ODS risk**: **Na ≤ 105**, **hypokalemia**, **alcohol use disorder**, **malnutrition**, **advanced liver disease**.

- **Osmotic demyelination syndrome (ODS)**: appears **2–6 days** after overly rapid correction → **dysarthria**, **dysphagia**, paraparesis/quadriparesis, behavioral changes, seizures, **"locked-in" syndrome** — often irreversible.
- **If overcorrected:** re-lower Na with **D5W** (e.g., 3 mL/kg/h) ± **desmopressin 2–4 mcg IV/SC every 6–8 h**.
- **Watch for auto-correction:** when the cause is removed (volume repletion, cortisol replacement, stopping thiazide/drug) → sudden water diuresis → rapid Na rise.
- **Potassium replacement also raises serum Na** (count K in correction calculations).

6. Hypernatremia and Hyperosmolar States

Overview

- **Na > 145 mEq/L • always hypertonic** • usually a **water deficit** (more water lost than sodium) • thirst normally prevents it → at risk: **elderly**, **infants**, **intubated/sedated**, altered mental status, no access to water.
- **Symptoms:** thirst, lethargy, irritability, weakness, confusion, hyperreflexia, **seizures**, coma • brain shrinkage → vascular rupture (intracranial hemorrhage).

Causes

Mechanism	Examples
Water loss (insensible/GI)	Fever, burns, sweating, mechanical ventilation, **osmotic diarrhea** (e.g., **lactulose**), vomiting, NG suction
Renal water loss — diabetes insipidus (DI)	**Central DI** (↓ ADH): head trauma, neurosurgery, tumors • **Nephrogenic DI** (kidney resistant to ADH): **lithium** (most common drug), **demeclocycline**, **amphotericin B**, foscarnet, **hypercalcemia**, **hypokalemia**
Osmotic diuresis	**Hyperglycemia** (HHS), **mannitol**, urea, high-protein tube feeds without enough free water
Sodium gain (less common)	**Hypertonic saline**, **sodium bicarbonate**, hypertonic tube feeds, salt ingestion, primary hyperaldosteronism

Treatment

- **Free water deficit (L) = TBW × [(serum Na ÷ 140) − 1]** • TBW = weight × 0.6 (men), 0.5 (women, elderly men), 0.45 (elderly women).
- **Example:** 70 kg man, Na 160 → 42 × (160/140 − 1) = **6 L** free water deficit.
- Replace the deficit over **48–72 hours** + **ongoing losses** + maintenance needs.
- **Correction rate (chronic): ≤ 10–12 mEq/L per 24 h** (~0.5 mEq/L/h) → too-fast correction causes **cerebral edema** and seizures • acute (< 48 h, e.g., salt poisoning) may be corrected faster (~1 mEq/L/h).
- **Fluid choice: hypovolemic/shock → NS first** to restore perfusion, then hypotonic fluid • **euvolemic → D5W** or **enteral free water** (preferred if gut works) • **hypervolemic (sodium gain) → D5W + loop diuretic** (dialysis if renal failure).
- **Monitor Na every 4–6 hours** during correction.

Diabetes insipidus treatment

Type	Treatment
Central DI	**Desmopressin (DDAVP)** — intranasal 10–40 mcg/day, oral 0.1–0.2 mg BID–TID, SC/IV 1–2 mcg BID • risk: **hyponatremia** (monitor Na, avoid excess water)
Nephrogenic DI	Treat cause (stop lithium if possible; correct Ca, K) • **thiazide diuretics** (paradoxically ↓ urine volume) • **amiloride** (lithium-induced) • **low-sodium, low-protein diet** • NSAIDs (indomethacin)

Other hyperosmolar states

- **Hyperosmolar hyperglycemic state (HHS):** glucose > 600, **serum osmolality > 320**, minimal ketones → **aggressive NS** first (large fluid deficit), switch to 0.45% NaCl when corrected Na is normal/high, add dextrose when glucose ~250–300, **insulin** only after **K ≥ 3.3**, replace potassium.
- **Mannitol** (osmotic diuretic for ↑ ICP): 0.25–1 g/kg IV • monitor serum osmolality/**osmolal gap** (hold if gap > 20 or osm > 320) • AKI, hypernatremia, volume depletion • use a filter (crystals).

7. Disorders of Potassium

Basics

- **Normal K: 3.5–5.0 mEq/L • 98% intracellular** → small serum changes = large total body changes.
- **Shift INTO cells (↓ serum K): insulin**, **beta-2 agonists** (albuterol), **alkalosis**, theophylline, caffeine, refeeding.
- **Shift OUT of cells (↑ serum K): acidosis**, insulin deficiency (DKA), **beta-blockers**, **digoxin toxicity** (Na/K-ATPase), **succinylcholine**, hyperosmolality, cell lysis.

Hypokalemia (K < 3.5)

Category	Causes
GI losses	**Diarrhea**, laxative abuse, vomiting/NG suction (mainly via metabolic alkalosis → renal K loss)
Renal losses — drugs	**Loop and thiazide diuretics** (most common), **amphotericin B**, **aminoglycosides**, **cisplatin**, foscarnet, high-dose **penicillins** (piperacillin), **corticosteroids**/fludrocortisone
Renal losses — other	**Hypomagnesemia** (K stays low until Mg is corrected), hyperaldosteronism, renal tubular acidosis, DKA treatment
Intracellular shift	**Insulin**, **albuterol**/beta-agonists, theophylline, caffeine, **alkalosis**, **refeeding syndrome**
Poor intake	Malnutrition, alcoholism (usually with other causes)

- **Severity:** mild 3.0–3.5 • moderate 2.5–3.0 • **severe < 2.5** or symptomatic.
- **Symptoms:** muscle **weakness**, cramps, fatigue, **constipation/ileus**, rhabdomyolysis, paralysis, **arrhythmias**.
- **ECG: flattened T waves**, **U waves**, ST depression, prolonged QT • ↑ **digoxin toxicity**.

Potassium replacement

- **Estimate:** each **10 mEq of KCl ↑ serum K by ~0.1 mEq/L** (normal renal function) • total deficit roughly **100–200 mEq** for each 1 mEq/L fall below 3.5 (more when K < 3).
- **Oral preferred** if K ≥ 2.5–3.0, asymptomatic, GI tract works • **40–100 mEq/day** in divided doses (≤ **20–40 mEq per dose** for tolerance).
- **KCl** = salt of choice (also corrects metabolic alkalosis) • **potassium citrate/bicarbonate** → with **metabolic acidosis** (RTA) • **potassium phosphate** → with **hypophosphatemia**.
- **Oral ADRs:** GI irritation, **ulceration** (ER tablets — swallow whole with food and a full glass of water, stay upright; liquid tastes bad).
- **Always correct magnesium** as well • reduce doses (~50%) and monitor closely in **renal impairment** • recheck K **2–4 h after IV** doses.

IV potassium safety (high-yield)	Limit
Peripheral line — max rate	**10 mEq/h** (some institutions up to 20 mEq/h)
Central line — max rate	**20 mEq/h** (up to 40 mEq/h only in emergencies with **continuous ECG monitoring**)
Peripheral concentration	~**10 mEq/100 mL** (≤ 80 mEq/L in maintenance fluids) → painful, phlebitis

IV potassium safety (high-yield)	Limit
Diluent	Prefer **NS** (avoid dextrose initially — dextrose → insulin → K shifts into cells)
NEVER	**Give IV push or undiluted → cardiac arrest** • KCl concentrate is a **high-alert medication** (use premixed bags)

Hyperkalemia (K > 5.0)

- **Rule out pseudohyperkalemia** first: **hemolyzed sample**, tight tourniquet/fist clenching, severe thrombocytosis or leukocytosis → repeat the test.

Category	Causes
↓ Renal excretion	**AKI/CKD** (most common), hypoaldosteronism, adrenal insufficiency
Drugs (high-yield)	**ACE inhibitors**, **ARBs**, aliskiren, **spironolactone**, **eplerenone**, **finerenone**, **amiloride**, **triamterene**, **NSAIDs**, **trimethoprim** (TMP-SMX), **pentamidine**, **heparin/LMWH** (↓ aldosterone), **cyclosporine**, **tacrolimus**, **drospirenone**, **beta-blockers**, **digoxin toxicity**, **succinylcholine**, K supplements, **salt substitutes (KCl)**
Shift out of cells	**Metabolic acidosis**, **DKA/insulin deficiency**, hyperosmolality, exercise
Cell breakdown / load	**Tumor lysis**, **rhabdomyolysis**, hemolysis, burns, GI bleeding, **blood transfusions**, PN

- **Symptoms:** muscle weakness, paresthesias, paralysis, **arrhythmias** (may be sudden, without warning).
- **ECG progression: peaked T waves** → PR prolongation and loss of P waves → **QRS widening** → sine wave → **VF/asystole**.

Treatment of hyperkalemia (3 steps)

Step	Drug	Dose	Onset / duration / notes
1. Stabilize the heart (ECG changes or K ≥ 6.5)	**Calcium gluconate 10%**	**1 g (10 mL) IV over 2–10 min**; repeat in 5 min if ECG unchanged (up to 3 g)	**1–3 min** / 30–60 min • **does NOT lower K** • peripheral OK
	Calcium chloride 10%	1 g IV (central line preferred)	**3× more elemental Ca** • **vesicant** (tissue necrosis) • cardiac arrest
2. Shift K into cells	**Regular insulin + dextrose**	**10 units IV** (5 units or 0.1 unit/kg in renal failure) + **25 g dextrose** (50 mL D50) if glucose < 250	**10–20 min** / 4–6 h • ↓ K 0.6–1 mEq/L • **monitor glucose hourly × 4–6 h** (hypoglycemia)
	Albuterol (nebulized)	**10–20 mg** (4–8× the asthma dose)	~30 min / 2–4 h • ↓ K 0.5–1 • tachycardia • adjunct to insulin
	Sodium bicarbonate	50–150 mEq IV	Only if **metabolic acidosis** (pH < 7.2) • not effective alone
3. Remove K from the body	**Loop diuretic** (furosemide)	40–80 mg IV	If kidneys work (+ fluids if volume-depleted)
	Sodium zirconium cyclosilicate (Lokelma)	**10 g TID** up to 48 h, then 5–10 g daily	**~1 h** • **edema** (sodium) • separate other oral drugs by **2 h**
	Patiromer (Veltassa)	8.4 g daily → max 25.2 g	**~7 h** • **hypomagnesemia**, constipation • separate other drugs by **3 h** • chronic use
	Sodium polystyrene sulfonate (Kayexalate)	15–30 g PO or 30–50 g PR	Slow, unreliable • **intestinal necrosis** (especially with sorbitol, post-op, ileus) • sodium load • separate drugs by 3 h
	Hemodialysis	—	**Most effective** • renal failure, life-threatening or refractory

- **Chronic hyperkalemia:** low-K diet (avoid salt substitutes), stop/reduce offending drugs, **loop or thiazide diuretics**, correct acidosis (sodium bicarbonate), **patiromer or SZC** (allow continued ACEi/ARB/MRA in HF and CKD), fludrocortisone (hypoaldosteronism).

8. Disorders of Magnesium Homeostasis

- **Normal Mg: 1.7–2.4 mg/dL** (≈ 1.4–2.0 mEq/L) • mostly in **bone and cells** → serum level may not reflect total body stores • regulated by the kidneys.
- **Conversion: 1 g magnesium sulfate = 8.1 mEq = 4 mmol = ~98.6 mg elemental Mg**.

Hypomagnesemia (Mg < 1.7 mg/dL)

Category	Causes
GI losses	**Diarrhea**, malabsorption, short bowel, NG suction, pancreatitis
Poor intake	**Alcohol use disorder** (very common), malnutrition, **refeeding syndrome**
Renal losses — drugs (high-yield)	**Loop and thiazide diuretics**, **proton pump inhibitors** (long-term; FDA warning), **aminoglycosides**, **amphotericin B**, **cisplatin**, **cyclosporine**, **tacrolimus**, **foscarnet**, pentamidine, **EGFR inhibitors** (cetuximab, panitumumab)
Other	DKA treatment, hungry bone syndrome, hyperaldosteronism, hypercalcemia

- **Symptoms:** neuromuscular irritability — **tremor**, muscle cramps, **tetany**, **Chvostek** and **Trousseau** signs, **seizures** • **arrhythmias** — QT prolongation, **torsades de pointes**, ↑ digoxin toxicity.
- **Causes refractory hypokalemia and hypocalcemia** (↓ PTH release and PTH resistance) → **correct Mg first**.

Magnesium replacement

Situation	Treatment
Mild, asymptomatic	**Oral**: **magnesium oxide 400 mg** (241 mg elemental) BID–TID • magnesium chloride or lactate (better absorbed, less diarrhea) • **diarrhea** limits oral dosing
Mild–moderate (IV)	**Magnesium sulfate 1–4 g IV** (e.g., 1–2 g over 1–2 h)
Severe (< 1 mg/dL) or symptomatic	**4–8 g IV over 12–24 h** • **slow infusion** (≤ ~1 g/h) — rapid infusion ↑ renal excretion (~50% lost)
Torsades de pointes / seizures	**1–2 g IV over 5–15 min**
Eclampsia / pre-eclampsia	4–6 g IV load, then 1–2 g/h

- **Renal impairment:** reduce dose (~50%) and monitor closely (accumulation) • recheck level 6–12 h after IV dose (slow equilibration) • rapid IV → flushing, hypotension.

Hypermagnesemia (Mg > 2.4 mg/dL)

- **Causes: renal failure** + Mg intake (**antacids**, **laxatives** — milk of magnesia, magnesium citrate, enemas), **IV magnesium** for pre-eclampsia, lithium, hypothyroidism, adrenal insufficiency, tumor lysis.
- **Symptoms by level:** ~4–6 mg/dL → nausea, flushing, lethargy, **↓ deep tendon reflexes** (first sign) • ~6–12 mg/dL → **hypotension**, **bradycardia**, ECG changes, somnolence • > 12–15 mg/dL → **respiratory paralysis**, complete heart block, **cardiac arrest**.
- **Treatment: stop magnesium • IV calcium gluconate 1–2 g** (antagonizes cardiac and neuromuscular effects) • **IV fluids + loop diuretic** (if renal function) • **hemodialysis** in renal failure or severe toxicity.

9. Disorders of Phosphorus Homeostasis

- **Normal phosphate: 2.5–4.5 mg/dL** • mostly in **bone** • main intracellular anion (ATP, 2,3-DPG).
- **Regulation: PTH** ↑ renal excretion • **vitamin D (calcitriol)** ↑ GI absorption • **FGF-23** ↑ renal excretion.
- **Dose IV phosphate in mmol** (mEq varies with pH).

Hypophosphatemia (< 2.5 mg/dL; severe < 1.0)

Category	Causes
Intracellular shift	**Refeeding syndrome** (most important), insulin, **DKA treatment**, **respiratory alkalosis**, hungry bone syndrome
↓ Intake/absorption	Malnutrition, **alcohol use disorder**, vitamin D deficiency, **phosphate binders** (calcium, aluminum, sevelamer), **Al/Mg antacids**, chronic diarrhea
↑ Renal loss	Hyperparathyroidism, **Fanconi syndrome** (**tenofovir**, ifosfamide), osmotic diuresis, **IV ferric carboxymaltose**, acetazolamide
Other	**Continuous renal replacement therapy (CRRT)**, burns, sepsis

- **Symptoms** (usually < 1 mg/dL): **muscle weakness**, **respiratory failure** (diaphragm weakness, **failure to wean** from ventilator), **rhabdomyolysis**, **hemolysis**, ↓ cardiac contractility/heart failure, confusion, seizures, impaired WBC function • chronic: **osteomalacia**.

Phosphate replacement

Severity	Treatment
Mild–moderate, asymptomatic (> 1–1.5 mg/dL)	**Oral**: **K-Phos Neutral** (250 mg = 8 mmol phosphorus per tablet) — **~0.5–1 g (16–32 mmol)/day** in divided doses • **diarrhea** limits dosing
Moderate (IV)	**0.16–0.32 mmol/kg** IV (e.g., 15–30 mmol) over 4–6 h
Severe (< 1 mg/dL) or symptomatic	**0.32–0.64 mmol/kg** IV (e.g., 30–45 mmol) over 4–6 h

- **Choose the salt by potassium level: potassium phosphate** (contains ~**4.4 mEq K per 3 mmol phosphate** → follow K infusion-rate limits) if K < 4 • **sodium phosphate** if K ≥ 4 or renal impairment.
- Use **ideal/adjusted body weight** in obesity • **reduce dose ~50% in renal impairment**.
- **Risks of rapid IV phosphate: hypocalcemia** (Ca-phosphate precipitation), metastatic calcification, **hyperphosphatemia**, **hyperkalemia** (K phosphate), AKI, arrhythmias • **do not infuse with calcium in the same line**.

Hyperphosphatemia (> 4.5 mg/dL)

- **Causes: CKD** (most common), AKI, **tumor lysis syndrome**, **rhabdomyolysis**, hemolysis, **hypoparathyroidism**, vitamin D toxicity, **sodium phosphate enemas/bowel preps** (acute phosphate nephropathy), excessive supplementation, DKA (early), lactic acidosis.
- **Effects:** mostly via **hypocalcemia** (tetany, seizures) and **calcification** — **Ca × Phos product > 55–60 mg²/dL²** → vascular/soft tissue calcification, calciphylaxis.
- **Treatment:** dietary phosphate restriction (800–1,000 mg/day) • **phosphate binders WITH meals** (calcium acetate/carbonate, sevelamer, lanthanum, ferric citrate, sucroferric oxyhydroxide; aluminum hydroxide ≤ 4 weeks only) • tenapanor • treat cause (TLS: hydration, allopurinol/rasburicase) • acute severe: IV saline (if renal function), **dialysis**.

10. Disorders of Calcium Homeostasis

Basics

- **Total calcium: 8.5–10.5 mg/dL • ionized calcium: 1.1–1.3 mmol/L** (4.5–5.3 mg/dL) — the active form • ~40% bound to **albumin**.
- **Corrected calcium** = measured Ca + **0.8 × (4 − albumin)** • **ionized Ca** is preferred in critically ill patients.
- **Alkalosis** ↑ protein binding → ↓ ionized Ca (tetany with hyperventilation) • acidosis ↑ ionized Ca.
- **Regulation: PTH** (↑ bone resorption, ↑ renal Ca reabsorption, ↑ calcitriol) • **calcitriol** (↑ GI absorption) • **calcitonin** (↓ bone resorption).

Hypocalcemia (corrected Ca < 8.5 or ionized < 1.1)

Category	Causes
↓ PTH	**Post-thyroidectomy/parathyroidectomy** (hungry bone), autoimmune hypoparathyroidism, **hypomagnesemia**
↓ Vitamin D	Deficiency, **CKD** (↓ calcitriol + ↑ phosphate), liver disease
Drugs (high-yield)	**Bisphosphonates** (zoledronic acid), **denosumab** (severe in advanced CKD/dialysis — boxed), **cinacalcet**/etelcalcetide, **foscarnet**, **loop diuretics**, phosphate, calcitonin, cisplatin, **phenytoin/phenobarbital** (↓ vitamin D)
Other	**Acute pancreatitis**, sepsis, **massive transfusion** (citrate chelates Ca), tumor lysis, rhabdomyolysis (early)

- **Symptoms** (neuromuscular irritability): **perioral numbness**, **paresthesias** of fingers/toes, muscle cramps, **tetany**, carpopedal spasm, **Chvostek sign** (facial twitch when tapping facial nerve), **Trousseau sign** (carpal spasm with BP cuff inflated), **laryngospasm**, **seizures** • **QT prolongation**, hypotension, heart failure.

Calcium replacement

Product	Elemental calcium	Key points
Calcium gluconate 10% (1 g)	**93 mg (4.65 mEq)**	**Preferred IV** (peripheral OK) • acute symptomatic: **1–2 g IV over 10–20 min**, then infusion if needed (e.g., 0.5–1.5 mg/kg/h elemental)
Calcium chloride 10% (1 g)	**273 mg (13.6 mEq)** (~3× gluconate)	**Central line** (vesicant — tissue necrosis) • cardiac arrest, severe hyperkalemia
Calcium carbonate (oral)	**40%**	**Take with meals** (needs acid) • ↓ absorption with **PPIs** • constipation, gas • cheapest
Calcium citrate (oral)	**21%**	**Any time** (no acid needed) • preferred with **PPIs**, achlorhydria, bariatric surgery

- **Correct magnesium** first or at the same time.
- **IV incompatibilities:** do not mix with **bicarbonate** or **phosphate** (precipitates) • **ceftriaxone** + calcium in neonates → fatal precipitates (lung/kidney).
- Caution with **digoxin** (calcium ↑ digoxin arrhythmia risk) → give slowly with monitoring.
- **Oral calcium:** chronic/asymptomatic: 1–3 g elemental/day + **vitamin D** • take **≤ 500–600 mg elemental per dose** • **separate** from **levothyroxine** (4 h), **iron**, **fluoroquinolones**, **tetracyclines**, **bisphosphonates**.
- **Vitamin D: cholecalciferol (D3)**/ergocalciferol for deficiency • **calcitriol** (active) for **hypoparathyroidism** and **CKD** • hypoparathyroidism: calcitriol + calcium, thiazide (↓ urinary Ca), recombinant PTH.
- **Massive transfusion:** give calcium to keep ionized Ca > 1.1 mmol/L.

Hypercalcemia (corrected Ca > 10.5)

- **Severity:** mild < 12 • moderate 12–14 • **severe > 14 mg/dL** (hypercalcemic crisis).
- **Causes: primary hyperparathyroidism** (most common outpatient) and **malignancy** (most common inpatient: PTHrP, bone metastases, calcitriol in lymphoma) → > 90% of cases • others: **vitamin D** or **vitamin A** toxicity, **thiazide diuretics**, **lithium**, **milk-alkali syndrome** (calcium carbonate + antacids), immobilization, **granulomatous disease** (sarcoidosis, TB — ↑ calcitriol), hyperthyroidism, adrenal insufficiency, **teriparatide/abaloparatide**, theophylline toxicity.
- **Symptoms — "stones, bones, abdominal groans, psychiatric moans": kidney stones**, **polyuria** (nephrogenic DI) → **dehydration**, AKI • bone pain, fractures • **constipation**, nausea, anorexia, pancreatitis, peptic ulcer • fatigue, **confusion**, depression, lethargy, coma • **short QT**, bradyarrhythmias • ↑ **digoxin toxicity**.

Treatment of hypercalcemia

Step / drug	Dose	Key points
Mild (< 12), asymptomatic	—	Hydration • avoid thiazides, lithium, calcium and vitamin D supplements • treat cause
1. IV isotonic saline (first step)	e.g., 1–2 L bolus, then **200–300 mL/h** → urine output 100–150 mL/h	Restores volume and ↑ renal Ca excretion • caution heart failure/renal failure
2. Calcitonin	**4 units/kg IM/SC every 12 h** (↑ to 8 units/kg q6–12h)	**Fastest** (4–6 h) but modest (↓ 1–2 mg/dL) • **tachyphylaxis** within 48 h → use only 24–48 h • nasal form not effective • nausea, flushing • salmon allergy
3. Bisphosphonate	**Zoledronic acid 4 mg IV over ≥ 15 min** (preferred, most potent) • **pamidronate 60–90 mg IV over 2–24 h**	Onset **2–4 days**, lasts weeks • hydrate first • **acute-phase reaction** (fever, flu-like), **nephrotoxicity** (adjust/avoid if CrCl < 30–35), hypocalcemia, **osteonecrosis of the jaw**, atypical femur fracture
Denosumab	120 mg SC	Bisphosphonate-refractory or **renal impairment** • hypocalcemia risk
Glucocorticoids	Prednisone 20–40 mg/day (or hydrocortisone IV)	For **calcitriol-mediated** hypercalcemia: **lymphoma**, **granulomatous disease**, vitamin D toxicity
Loop diuretics	Furosemide IV	**Only after volume repletion** and for fluid overload — not routine
Hemodialysis	—	Severe hypercalcemia with renal failure or heart failure
Cinacalcet / surgery	30 mg BID → titrate	Primary hyperparathyroidism (parathyroidectomy is definitive) • parathyroid carcinoma

11. Fluids and Electrolytes Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
0.9% NaCl	Resuscitation: 30 mL/kg (sepsis) or 500–1,000 mL boluses	**Hyperchloremic acidosis**, volume overload	Preferred in TBI, hypochloremic alkalosis
Lactated Ringer's	Same as NS	Volume overload	Contains Ca (not with blood/ceftriaxone in neonates) • avoid in TBI
Plasma-Lyte A	Same	Volume overload	Balanced, Ca-free
D5W	Per free water deficit	**Hyponatremia**, hyperglycemia	Never for resuscitation
0.45% NaCl	Per need	Hyponatremia	Hypotonic
3% NaCl	100–150 mL over 10–20 min (repeat ×2–3) or slow infusion	**ODS** (overcorrection), volume overload	Na every 2–4 h • limit 8–10 mEq/L/24 h
23.4% NaCl	30 mL over 10–20 min (central) for ICP	Hypernatremia, phlebitis	**High-alert** • dilute for additives
Albumin 5% / 25%	5%: 250–500 mL • 25%: 25–100 g	Volume overload	Avoid in TBI • expensive
Tolvaptan	15 mg daily → 60 mg (≤ 30 days)	**ODS**, **hepatotoxicity** (boxed), thirst, polyuria	**Start in hospital** • no fluid restriction first 24 h • CI strong CYP3A4 inhibitors
Conivaptan	20 mg IV load → 20–40 mg/day × ≤ 4 days	Infusion-site reactions, hypotension	**Strong CYP3A4 inhibitor**

Drug	Dose	Key ADRs	Key notes / interactions
Urea (oral)	15–60 g/day	Bad taste, GI upset	SIADH
Desmopressin	Nasal 10–40 mcg/day • PO 0.1–0.2 mg BID–TID • IV 1–2 mcg	**Hyponatremia**	Central DI • re-lowering Na after overcorrection
Potassium chloride (oral)	40–100 mEq/day divided (≤ 20–40 mEq/dose)	GI irritation/ulceration	Swallow ER whole with food and water
Potassium chloride (IV)	Peripheral ≤ 10 mEq/h • central ≤ 20 mEq/h	Phlebitis, **hyperkalemia**, arrhythmia	**Never IV push** • NS diluent • ↓ dose in renal impairment
Potassium citrate	10–20 mEq BID–TID	GI upset	Hypokalemia + acidosis (RTA), kidney stones
Calcium gluconate	1–2 g IV over 10–20 min (hyperK: 1 g over 2–10 min)	Bradycardia, hypotension (fast), extravasation	Not with bicarbonate/phosphate • ceftriaxone in neonates • digoxin caution
Calcium chloride	1 g IV (central)	**Vesicant**	3× elemental Ca
Regular insulin + dextrose	10 units IV + 25 g dextrose	**Hypoglycemia**	Glucose hourly × 4–6 h
Albuterol (nebulized)	10–20 mg	Tachycardia, tremor	Adjunct for hyperkalemia
Sodium bicarbonate	50–150 mEq IV	Sodium load, alkalosis, hypocalcemia	Only with acidosis
Sodium zirconium cyclosilicate	10 g TID × 48 h → 5–10 g daily	**Edema**	Separate drugs 2 h
Patiromer	8.4–25.2 g daily	**Hypomagnesemia**, constipation	Separate drugs 3 h
Sodium polystyrene sulfonate	15–30 g PO / 30–50 g PR	**Intestinal necrosis**, sodium load	Separate drugs 3 h
Furosemide	40–80 mg IV	Hypokalemia, hypomagnesemia, ototoxicity	K removal, fluid overload
Magnesium oxide	400 mg BID–TID	**Diarrhea**	Separate from quinolones, tetracyclines
Magnesium sulfate IV	1–4 g (severe 4–8 g over 12–24 h; torsades 1–2 g over 5–15 min)	Flushing, hypotension, ↓ **reflexes** (toxicity)	↓ dose 50% in renal impairment • antidote calcium gluconate
Potassium phosphate IV	0.16–0.64 mmol/kg over 4–6 h	**Hyperkalemia**, hypocalcemia, hyperphosphatemia	~4.4 mEq K per 3 mmol • use if K < 4
Sodium phosphate IV	0.16–0.64 mmol/kg over 4–6 h	Hypocalcemia, hyperphosphatemia	Use if K ≥ 4 or renal impairment
K-Phos Neutral (oral)	1–2 tablets BID–QID (8 mmol each)	**Diarrhea**	Separate from Ca, Mg, Al
Calcium carbonate	500–1,500 mg elemental/day (with meals)	Constipation, gas, milk-alkali	Needs acid (PPIs ↓) • chelates many drugs
Calcium citrate	500–1,500 mg elemental/day (any time)	GI upset	OK with PPIs
Calcitriol	0.25–0.5 mcg daily	**Hypercalcemia**, hyperphosphatemia	Hypoparathyroidism, CKD
Cholecalciferol	1,000–2,000 IU daily (higher for deficiency)	Hypercalcemia (toxicity)	Vitamin D deficiency
Calcitonin	4–8 units/kg IM/SC q6–12h	Nausea, flushing	**Tachyphylaxis** 48 h
Zoledronic acid	4 mg IV over ≥ 15 min	Acute-phase reaction, **nephrotoxicity**, hypocalcemia, ONJ	Hydrate first • renal adjust

Drug	Dose	Key ADRs	Key notes / interactions
Pamidronate	60–90 mg IV over 2–24 h	Same	Same
Denosumab	120 mg SC	**Hypocalcemia**, infections, ONJ	Renal impairment option
Prednisone	20–40 mg/day	Hyperglycemia, infection	Calcitriol-mediated hypercalcemia
Cinacalcet	30 mg BID → titrate	**Hypocalcemia**, nausea	**CYP2D6 inhibitor** • 3A4 substrate

Exam Traps and Frequently Tested Points

1. **TBW = 60% (men), 50% (women)** • 2/3 ICF, 1/3 ECF • maintenance **30–35 mL/kg/day** or **4-2-1** rule.
2. **NS: Na 154, Cl 154, 308 mOsm** → hyperchloremic acidosis • **LR: Na 130, K 4, Ca 3, lactate 28** • balanced crystalloids preferred in most ICU patients.
3. **NS preferred in brain injury** and hypochloremic alkalosis • **no albumin in TBI** • **avoid hydroxyethyl starch**.
4. 1 L crystalloid → **~250 mL** stays intravascular • 1 L D5W → **~80 mL**.
5. **Sepsis: 30 mL/kg** crystalloid in 3 h.
6. **Serum osm = 2Na + glucose/18 + BUN/2.8** (275–295) • **osmolal gap > 10** → methanol, ethylene glycol, propylene glycol, mannitol.
7. **Corrected Na** = Na + 1.6 per 100 mg/dL glucose above 100.
8. **Peripheral line osmolarity ≤ 900 mOsm/L**.
9. **Never give sterile water IV** (hemolysis) • hypotonic fluids → hospital-acquired hyponatremia (children, post-op).
10. **Hyponatremia:** check osmolality → volume status → urine Na/osm.
11. **Thiazides** → hyponatremia • **SIADH drugs:** SSRIs, carbamazepine/oxcarbazepine, cyclophosphamide, vincristine, chlorpropamide, MDMA, desmopressin.
12. **SIADH:** euvolemic, urine osm > 100, urine Na > 30 → **fluid restriction** first.
13. **Severe symptomatic hyponatremia → 3% NaCl 100–150 mL bolus** (↑ Na 4–6 mEq/L).
14. **Correction limit: ≤ 8–10 mEq/L per 24 h** (≤ 8 if high ODS risk: Na ≤ 105, alcoholism, malnutrition, hypokalemia, liver disease).
15. **Overcorrection → D5W ± desmopressin** • **ODS** = dysarthria, dysphagia, paralysis, locked-in.
16. **Tolvaptan:** start in hospital, no fluid restriction 24 h, avoid liver disease, ≤ 30 days • **conivaptan** = strong CYP3A4 inhibitor.
17. **Hypernatremia:** free water deficit = **TBW × (Na/140 − 1)** • correct ≤ **10–12 mEq/L/24 h** • shock → NS first.
18. **Lithium, demeclocycline, amphotericin** → nephrogenic DI • central DI → **desmopressin** • nephrogenic → thiazide, amiloride (lithium).
19. **Hypokalemia:** correct **Mg first** • ECG U waves, flat T • 10 mEq KCl ↑ K ~0.1.
20. **IV KCl: peripheral ≤ 10 mEq/h, central ≤ 20 mEq/h**, **never IV push**, avoid dextrose diluent.
21. **Hyperkalemia drugs:** ACEi/ARB, spironolactone, trimethoprim, NSAIDs, heparin, tacrolimus, cyclosporine, succinylcholine, digoxin toxicity.
22. **Hyperkalemia steps: calcium gluconate** (stabilize, doesn't lower K) → **insulin + dextrose**, albuterol, (bicarbonate if acidosis) → **remove**: loop diuretic, SZC, patiromer, SPS, dialysis.
23. **Peaked T waves** = hyperkalemia • rule out **hemolyzed sample**.
24. **SZC** → fastest binder (1 h), edema • **patiromer** → hypomagnesemia, separate 3 h • **SPS** → intestinal necrosis.
25. **1 g MgSO4 = 8.1 mEq = 4 mmol** • infuse slowly (≤ 1 g/h) • torsades → 1–2 g IV.
26. **Hypomagnesemia drugs: PPIs**, diuretics, cisplatin, amphotericin, aminoglycosides, cyclosporine/tacrolimus, cetuximab.
27. **Hypermagnesemia:** first sign **loss of deep tendon reflexes** → antidote **calcium gluconate**.
28. **Hypophosphatemia: refeeding syndrome**, alcoholism, DKA treatment, binders, IV ferric carboxymaltose → weakness, **respiratory failure**, hemolysis.
29. **K phosphate** if K < 4 (4.4 mEq K per 3 mmol) • **Na phosphate** if K ≥ 4 • dose in **mmol**.

30. **Corrected Ca = Ca + 0.8 × (4 − albumin)** • ionized Ca best in ICU.
31. **Hypocalcemia:** Chvostek, Trousseau, tetany, ↑ QT • **calcium gluconate IV** (chloride = 3× Ca, central, vesicant) • correct Mg.
32. **Calcium carbonate** 40% with meals (PPIs ↓) • **calcium citrate** 21%, any time.
33. **Hypercalcemia:** hyperparathyroidism and malignancy > 90% • stones, bones, groans, moans • short QT.
34. **Hypercalcemia treatment: NS first → calcitonin** (fast, tachyphylaxis) + **zoledronic acid** (2–4 days) • steroids for lymphoma/sarcoid • loop diuretics only after rehydration.
35. **Thiazides, lithium, vitamin D/A, calcium + antacids (milk-alkali)** → hypercalcemia.

> **Note:** Normal ranges, rates, and doses vary slightly between references and institutions. **Use the numbers in the reference you are studying for the exam.**

Good luck on your exam!